

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Trichloroacetic acid

**CAS No** 76-03-9

**Synonyms** TCA; Trichloroethanoic acid (Crystalline/Flakes/Certified/Certified ACS)

**Product Code** A322-3; A322-100; A322-500; A323-500; A324-3; A324-500; BP555-1; BP555-250; BP555-500; SA433-500

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** CHEMTREC®  
03 9757 4559 or +613 9757 4559

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
No hazards identified

### **Health hazards**

|                                                    |              |
|----------------------------------------------------|--------------|
| Skin Corrosion/Irritation                          | Category 1 A |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Specific target organ toxicity - (single exposure) | Category 3   |

### **Environmental hazards**

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

### **Label Elements**



Exclamation Mark



Corrosion



Environment

**Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage

H335 - May cause respiratory irritation

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P363 - Wash contaminated clothing before reuse

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component            | CAS No  | Weight % |
|----------------------|---------|----------|
| Trichloroacetic acid | 76-03-9 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician or poison control center immediately. If not breathing, give artificial respiration.

**Ingestion**

Do NOT induce vomiting. Immediate medical attention is required. Never give anything by mouth to an unconscious person. Drink plenty of water.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                             |
| <b>General Advice</b>                      | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Chloroform, Carbon dioxide (CO<sub>2</sub>), Phosgene, Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas.

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component            | Australia                                | New Zealand WEL                          | ACGIH TLV    | The United Kingdom | Germany                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|------------------------------------------|------------------------------------------|--------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroacetic acid | TWA: 1 ppm<br>TWA: 6.7 mg/m <sup>3</sup> | TWA: 1 ppm<br>TWA: 6.7 mg/m <sup>3</sup> | TWA: 0.5 ppm |                    | TWA: 0.2 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 1.4 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 0.2 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 1.4 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 0.2 ppm<br>Höhepunkt: 1.4 mg/m <sup>3</sup> |

### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Butyl rubber   | > 480 minutes     | 0.7 mm          | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Particulates filter conforming to EN 143 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                 |                                          |
|------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                           |                                          |
| <b>Physical State</b>                          | Solid                           |                                          |
| <b>Odor</b>                                    | of vinegar                      |                                          |
| <b>Odor Threshold</b>                          | No data available               |                                          |
| <b>pH</b>                                      | 1.2                             | (0.1M)                                   |
| <b>Melting Point/Range</b>                     | 52 - 58 °C / 125.6 - 136.4 °F   |                                          |
| <b>Softening Point</b>                         | No data available               |                                          |
| <b>Boiling Point/Range</b>                     | 196 °C / 384.8 °F               | @ 760 mmHg                               |
| <b>Flash Point</b>                             | No information available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available        |                                          |
| <b>Explosion Limits</b>                        | No data available               |                                          |
| <b>Vapor Pressure</b>                          | 1.2 mbar @ 50°C, 0.08 mbar @25C |                                          |
| <b>Vapor Density</b>                           | Not applicable                  | Solid                                    |
| <b>Specific Gravity / Density</b>              | 1.620                           |                                          |
| <b>Bulk Density</b>                            | No data available               |                                          |
| <b>Water Solubility</b>                        | 120 g/100 mL (20°C)             |                                          |
| <b>Solubility in other solvents</b>            | No information available        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                  |                                          |
| Trichloroacetic acid                           | 1,44                            |                                          |
| <b>Autoignition Temperature</b>                | No data available               |                                          |
| <b>Decomposition Temperature</b>               | No data available               |                                          |
| <b>Viscosity</b>                               | Not applicable                  | Solid                                    |
| <b>Explosive Properties</b>                    | No information available        |                                          |
| <b>Oxidizing Properties</b>                    | No information available        |                                          |
| <b>Other information</b>                       |                                 |                                          |
| <b>Molecular Formula</b>                       | C2 H Cl3 O2                     |                                          |
| <b>Molecular Weight</b>                        | 163.39                          |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                                                                                |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat.                                                                                                                       |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Metals.                                                                                                                   |
| <b>Hazardous Decomposition Products</b> | Chloroform. Carbon dioxide (CO <sub>2</sub> ). Phosgene. Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas. |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                  |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Based on available data, the classification criteria are not met |
| <b>Dermal</b>              | No data available                                                |
| <b>Inhalation</b>          | No data available                                                |

| Component            | LD50 Oral      | LD50 Dermal               | LC50 Inhalation |
|----------------------|----------------|---------------------------|-----------------|
| Trichloroacetic acid | 3320 mg/kg rat | LD50 > 2000 mg/kg ( Rat ) |                 |

**(b) skin corrosion/irritation;** Category 1 A

**(c) serious eye damage/irritation;** Category 1

|                                               |                   |
|-----------------------------------------------|-------------------|
| <b>(d) respiratory or skin sensitization;</b> |                   |
| <b>Respiratory</b>                            | No data available |
| <b>Skin</b>                                   | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;**

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component            | Australia | New Zealand          | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|----------------------|-----------|----------------------|-----------------|-------------------|----------|----|----|---------|
| Trichloroacetic acid |           | Suspected carcinogen |                 |                   | Group 2B |    |    |         |

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** Category 3

**Results / Target organs** Respiratory system

**(i) STOT-repeated exposure;** Based on available data, the classification criteria are not met

**Test method**  
**Test species / Duration**  
**Study result**  
**Route of exposure**  
**Target Organs**

Chronic Toxicity  
 dog / 90 days  
 NOEL = 26 mg/kg/d  
 Oral  
 None known.

**(j) aspiration hazard;**  
 Not applicable  
 Solid

**Symptoms / effects, both acute and delayed**  
 Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

**Ecotoxicity effects**  
 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Fish | Water Flea | Freshwater Algae | Microtox |
|----------------------|-----------------|------------|------------------|----------|
| Trichloroacetic acid | >277 mg/l       | 110 mg/l   | 0.27 mg/l        |          |

**Persistence and Degradability**  
**Persistence**  
 Not readily biodegradable  
**Degradation in sewage treatment plant**  
 Soluble in water, Persistence is unlikely, based on information available. Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential**  
 Product has a low potential to bioconcentrate Bioaccumulation is unlikely

| Component            | log Pow | Bioconcentration factor (BCF) |
|----------------------|---------|-------------------------------|
| Trichloroacetic acid | 1,44    | 0.4-1.7 Cyprinus caprio       |

**Mobility**  
 The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility Highly mobile in soils  
**Endocrine Disruptor Information**  
 This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant**  
 This product does not contain any known or suspected substance  
**Ozone Depletion Potential**  
 This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products**  
 Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**  
 Dispose of this container to hazardous or special waste collection point.

**Other Information**  
 Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge. Do not let this chemical enter the environment.

## Section 14 - Transport Information

### IMDG/IMO

**UN-No**  
**Proper Shipping Name**  
**Hazard Class**  
**Packing Group**

UN1839  
 Trichloroacetic acid, solid  
 8  
 II

**ADG**

UN-No UN1839  
 Proper Shipping Name Trichloroacetic acid, solid  
 Hazard Class 8  
 Packing Group II

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Trichloroacetic acid<br>76-03-9 ( >95 ) | 2X           |

**IATA**

UN-No UN1839  
 Proper Shipping Name Trichloroacetic acid  
 Hazard Class 8  
 Packing Group II

**Environmental hazards** Dangerous for the environment  
 Product is a marine pollutant according to the criteria set by IMDG/IMO

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                      | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                             |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trichloroacetic acid - 76-03-9 | Schedule 4 listed - for human dermal use except when in preparations containing <=12.5% of Trichloroacetic acid for the treatment of warts other than anogenital warts<br>Schedule 6 listed - except when included in Schedule 4 or 5, or in human dermal preparations containing <=12.5% of Trichloroacetic acid for the treatment of warts other than anogenital warts |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                      | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------------|-------------------------------------------------------------|------------------------|
| Trichloroacetic acid - 76-03-9 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable



**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

| Component                      | Australia | New South Wales | Western Australia | New Zealand          |
|--------------------------------|-----------|-----------------|-------------------|----------------------|
| Trichloroacetic acid - 76-03-9 |           |                 |                   | Suspected carcinogen |

**International Inventories**

| Component            | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|----------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Trichloroacetic acid | X    | X     | 200-927-2 | -      | X    | X   | -    | X     | X    | X    | X     | KE-34058 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component            | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Trichloroacetic acid | 76-03-9 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component            | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trichloroacetic acid | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances

## Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**MARPOL** - International Convention for the Prevention of Pollution from Ships**NZS 5433:2012** - Transport of Dangerous Goods on Land**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**WEL** - Workplace Exposure Limit**DNEL** - Derived No Effect Level**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**VOC** - (Volatile Organic Compound)**KECL** - Korean Existing and Evaluated Chemical Substances**CAS** - Chemical Abstracts Service**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail**OECD** - Organisation for Economic Co-operation and Development**LC50** - Lethal Concentration 50%**ATE** - Acute Toxicity Estimate**RPE** - Respiratory Protective Equipment**NOEC** - No Observed Effect Concentration**BCF** - Bioconcentration factor**PBT** - Persistent, Bioaccumulative, Toxic

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical incident response training.

## Revision Date

21-Nov-2022

## Revision Summary

Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet